

Flying Iron

Technical Data / Extra Information Datasheet

Generic template for theatre lifting, flying and suspension applications

Document control and scope

Document status	Generic guidance template - complete site-specific fields before relying on it as a controlled record.
Equipment category	Lifting Equipment / Lifting Accessory
Portal equipment item	Flying Iron
Primary use	Connection or attachment point for flown scenery, bars, cloths or other approved suspended items.
Prepared for	Stage Electrics Mechanical Theatre Equipment Compliance Portal

Important limitation

This datasheet is not a manufacturer manual or proof of rating. A flying iron must only be used where its rating, configuration, material condition and method of attachment are known and suitable for the intended load path.

1. Equipment overview

A flying iron is a metal attachment or bracket used within a theatre flying or suspension arrangement to connect a load to a line, bar, scenery component, suspension fitting or associated lifting accessory. It may form part of the load path between the lifted item and the lifting or support system.

- It should be treated as a load-bearing component, not merely as a general fixing.
- The complete load path should include the flying iron, fasteners, rope or wire connection, shackles, parent structure, scenery construction and any supporting bar or lifting system.
- The item should only be used where the WLL/SWL or design capacity is known, traceable and appropriate for the application.
- If there is uncertainty over rating, material, weld condition, method of attachment or loading direction, the item should be removed from service pending competent assessment.

2. Technical data record

Asset and configuration details	
Manufacturer / supplier	To be completed
Model / type / pattern	To be completed
Asset ID / reference	To be completed
Serial / batch / traceability mark	To be completed
Material / finish	To be completed - for example steel, painted, galvanised or stainless finish.
Rated capacity / WLL / SWL	To be completed - include units and configuration.
Approved load direction	To be completed - for example straight-line pull, vertical suspension, paired use or specific angle limits.
Compatible connection method	To be completed - shackle size, rope/wire termination, bolt size, hole diameter or other fitting.
Parent structure / item attached to	To be completed - scenery frame, bar, wire rope line, clew, chain, bracket or other structure.
Installation / fixing method	To be completed - bolts, welds, clamps, shackles, pins or other approved method.
Inspection interval	To be completed - in line with site inspection regime, manufacturer guidance and competent person requirements.
Known limitations	To be completed - configuration, angle, shock load, side load, environment, temperature, corrosion or use restrictions.

3. Intended use and limitations

- Use only as part of a planned lifting, flying or suspension arrangement assessed by a competent person.
- Do not exceed the marked or verified WLL/SWL, including any reduced capacity due to angle, connection method or paired/multiple component configuration.
- Do not use if the flying iron is homemade, unmarked, unidentified or has no suitable evidence of rating unless a competent person has verified its suitability.
- Do not use for side loading, twisting, prying or shock loading unless specifically designed and rated for that condition.

- Do not drill, weld, grind, heat, bend, reshape or repair without competent design approval and subsequent inspection.
- Do not rely on timber, scenic fabric, thin plate or unverified scenic construction as a structural parent component unless the complete assembly has been assessed.

4. Compatibility and installation considerations

Area	What must be confirmed	Notes / evidence required
Rating	WLL/SWL is known, legible and appropriate.	Rating plate, marking, certificate, drawing, manufacturer information or competent person assessment.
Load direction	The load is applied in the intended direction.	Avoid twist, side load and eccentric loading unless specifically approved.
Connections	Shackles, pins, bolts, links or rope terminations fit correctly.	No forcing, misalignment, undersized pins or poor bearing surfaces.
Parent item	The scenery, bar, frame or structure can transfer the load safely.	Check material, welds, joints, fixings, timber strength or structural design.
Fasteners	Bolts, nuts, washers, pins and retainers are correct and secure.	Use specified grade and locking method; do not substitute unknown hardware.
Access	Inspection, fitting and removal can be carried out safely.	Avoid hidden load-bearing fixings that cannot be inspected.

5. Pre-use and periodic inspection checks

- Confirm the item is the correct flying iron for the intended load, connection and configuration.
- Check that WLL/SWL, identification or traceability information is present and legible where applicable.
- Inspect the body for cracks, distortion, bending, elongation of holes, corrosion, wear, impact damage or sharp edges.
- Inspect welds, bends, eyes, holes, attachment plates and load-bearing edges for cracking or deformation.
- Check that all bolts, nuts, pins, shackles, washers and retainers are present, correct, tight and compatible.
- Check for signs of unauthorised drilling, grinding, welding, heating, reshaping or repair.
- Confirm the parent scenery, bar, frame or structural item is strong enough and free from damage at the connection point.
- Check that the flying iron is loaded in the intended direction and is not subject to twisting, leverage or side loading.
- Remove from service if rating, condition, compatibility or installation method is uncertain.

6. Defects and removal from service

Condition found	Immediate action	Typical follow-up
Crack, distortion or hole elongation	Remove from service immediately.	Competent assessment. Do not repair unless specifically approved.
Unknown or missing rating	Do not use for lifting or suspension.	Verify rating, replace, or have suitability assessed.
Corrosion, wear or sharp damage	Quarantine if capacity or fit may be affected.	Inspect, clean, replace or obtain competent advice.
Incorrect bolts, pins or shackles	Stop use and correct before loading.	Replace with specified compatible hardware.
Unauthorised modification or repair	Remove from service.	Review design and records before any return to service.
Parent structure damage	Do not load the assembly.	Assess scenery, bar, frame or support before continued use.

7. Regulatory and good-practice context

Guidance summary

A flying iron can be part of lifting equipment, a lifting accessory or a load-bearing support depending on how it is used. The actual legal duties depend on the installation, workplace use and risk assessment.

Reference	Relevance to flying irons	Practical implication
LOLER 1998	Relevant where the flying iron forms part of lifting equipment, a lifting accessory or a load path used for lifting or lowering loads.	Strength, stability, marking, planning and thorough examination requirements may apply.
PUWER 1998	Relevant because the item is work equipment used in a workplace or venue environment.	It must be suitable, maintained, inspected where deterioration creates risk, and used by trained persons.
BS EN 17206	Relevant where the flying iron forms part of entertainment machinery, stage lifting or production-area load-bearing systems.	Compatibility, documentation, inspection and control of load-bearing components should be considered.
ABTT Technical Standards	Provides recognised theatre-industry good-practice context.	Supports competent inspection, record keeping and venue-specific assessment.

8. Records to retain

- Manufacturer data, drawing, certificate or competent person statement confirming rating and intended use.
- Asset identification or location record for site-specific flying irons.
- Inspection and thorough examination records where required by the site regime or competent person.
- Records of any removal from service, defect, repair, modification or replacement.
- Evidence that connected components, such as shackles, slings, ropes, pins or parent structures, are also suitable and inspected.

9. Site-specific completion box

To be completed before use as a controlled record	
Venue / site	
Location / system reference	
Normal use	
Approved loading arrangement	
Restrictions / exclusions	
Inspection frequency	
Completed by / date	

10. Portal upload information

Upload the PDF version to the portal documents folder using this path:

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documents/flying_iron_technical_datasheet.pdf
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When the portal JavaScript is updated, use this document record for the Flying Iron equipment item:

```
documents: [  
  {  
    title: "Flying Iron Technical Datasheet",  
    file: "documents/flying_iron_technical_datasheet.pdf"  
  }  
]
```

Final note

This record should be reviewed against the actual item, manufacturer documentation and the venue inspection regime before being treated as controlled technical data.